

On the Anatomy of Trend

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Abstract

We present a theoretical framework for analyzing the performance of time-series momentum (trend-following) strategies when the underlying asset follows a general stationary Gaussian process. We derive closed-form expressions for the unconditional expected log payoff, decomposing the performance into a drift component related to the expected position and a timing component related to the alignment of the trend signal and the subsequent return of the underlying. Extending the analysis to conditional performance, we show that the conditional expected payoff preserves the characteristics from the unconditional: a conditional expected position component and a conditional timing component. We derive exact asymptotic limits for the conditional payoff in “deep crash” scenarios over a fixed horizon and show that even with i.i.d. returns, the conditional expected return is positive if the drawdown period is larger than one day. Finally, we show that a sufficiently long and severe drawdown always yields positive trend performance, irrespective of the underlying autocorrelation structure. We also characterize the role of the lookback parameter, revealing a trade-off between expected returns and hedge effectiveness. Empirical analysis of 103 assets over 25 years reveals an asymmetry in the evidence: the theory’s predictions about the *shape* of the payoff a strongly convex conditional profile (“smile”), robust hedging in tail events are confirmed with high statistical significance, while the unconditional timing benefit is economically positive but statistically fragile. The significant negative intercept (“cost of convexity”) completes the picture of trend-following as an insurance-like strategy: the convex payoff and downside protection are reliable, but they come at a quantifiable premium.

1 Introduction

Time-series momentum, or trend-following, has been a cornerstone of systematic trading for decades (Moskowitz et al., 2012; Hurst et al., 2017; Lempérière et al., 2014). Building on the foundational work on cross-sectional momentum (Jegadeesh and Titman, 1993) and its extension to time-series strategies across asset classes (Asness et al., 2013), this paper addresses some fundamental questions for this widespread hedge fund strategy: What drives trend returns? How does trend performance

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depend upon realized market outcomes? How does trend perform during drawdowns? And how do these properties depend on the underlying return dynamics? Basically, what do you need to believe in order to believe in trend as a return generator and hedge provider?

This paper provides a theoretical framework for trend-following strategies under general stationary Gaussian processes. We derive closed-form expressions for both unconditional and conditional expected payoffs, decomposing performance into intuitive drift and timing components. Our approach complements the analytical framework of [Grebencov and Serror \(2014\)](#), who derived the full unconditional P&L distribution (including variance, skewness, kurtosis, and quantiles) for exponential moving average strategies under a Gaussian model. We extend this line of work in three directions: (i) conditional analysis given realized benchmark performance, (ii) exact asymptotic characterization of hedging in crash scenarios, and (iii) a threshold property guaranteeing profitable trend-following in sufficiently severe drawdowns.

Our main contributions are fourfold. First, we show that unconditional and conditional expected returns share the same two-component structure: an average directional exposure to the underlying drift and a timing component coming from the alignment between the signal and subsequent returns. This decomposition provides theoretical grounding for the empirically observed convex (“option-like”) payoff profile of trend strategies documented by [Fung and Hsieh \(2001\)](#). Second, we establish exact asymptotic limits for trend performance in deep crash scenarios over fixed horizons, demonstrating that the strategy acts as a near-perfect hedge when signals track the drawdown. Third, we prove that for sufficiently severe linear drawdowns over long horizons, trend strategies are always profitable even when the underlying process exhibits mean reversion. This “threshold property” reveals that the directional trend component eventually dominates any adverse timing effects, complementing empirical findings on momentum crashes and conditional performance ([Daniel and Moskowitz, 2016](#); [Barroso and Santa-Clara, 2015](#)). Fourth, we provide an empirical validation using 103 assets across four asset classes. The evidence reveals a clear hierarchy: the theory’s *conditional* predictions the convex payoff profile, the significant downside betas, the hedging in tail events are confirmed with strong statistical significance, whereas the *unconditional* timing benefit is economically positive but statistically uncertain. The significant negative alpha (“cost of convexity”) positions trend-following as an insurance-like strategy whose value derives primarily from its payoff shape rather than from standalone expected returns. We also document a trade-off in drawdown protection: short lookback windows are optimal for hedging shallow market corrections, whereas long lookback windows are better at capturing the profits from deep, prolonged crashes.

The paper proceeds as follows. Section 2 introduces the Gaussian framework and establishes notation. Section 3 presents key Gaussian half-space identities used throughout. Section 4 derives unconditional expected returns and their drift-timing decomposition. Section 5 extends the analysis to conditional returns given realized cumulative performance. Section 6 establishes asymptotic hedge properties in deep crash and linear drawdown regimes. Section 7 specializes the results to ARMA processes and provides numerical illustrations with MA(1) benchmarks. Section 8 tests

the theoretical predictions using a multi-asset dataset of daily returns. Technical derivations are collected in the Appendix.

2 Model

Let $(X_t)_{t \in \mathbb{Z}}$ denote one-period *log* returns of a benchmark. We assume:

- (X_t) is a *stationary Gaussian* process,
- with mean $\mathbb{E}[X_t] = \mu$, variance $\text{Var}(X_t) = \sigma^2$,
- and autocovariance function

$$\gamma(h) := \text{Cov}(X_t, X_{t+h}), \quad h \in \mathbb{Z},$$

independent of t .

In particular $X_t \sim \mathcal{N}(\mu, \sigma^2)$ for every t .

Fix a lookback $k \geq 1$ and define the k -period signal and binary position

$$S_t := \sum_{j=1}^k X_{t-j}, \quad D_t := \text{sign}(S_t) \in \{-1, +1\}. \quad (1)$$

We study the per-period *log payoff*

$$Y_t := D_t X_t. \quad (2)$$

The signal is Gaussian $S_t \sim \mathcal{N}(k\mu, \sigma_S^2)$ with variance

$$\sigma_S^2 = \text{Var}(S_t) = \sum_{j=1}^k \sum_{\ell=1}^k \gamma(j - \ell). \quad (3)$$

The covariance c_{sx} follows directly from linearity of covariance and stationarity:

$$c_{sx} = \text{Cov}\left(\sum_{j=1}^k X_{t-j}, X_t\right) = \sum_{j=1}^k \text{Cov}(X_{t-j}, X_t) = \sum_{j=1}^k \gamma(j). \quad (4)$$

The unconditional average position of the strategy is

$$\mathbb{E}[D_t] = 2 \Phi\left(\frac{m_s}{\sqrt{v_s}}\right) - 1 = 2 \Phi\left(\frac{k\mu}{\sigma_S}\right) - 1. \quad (5)$$

For a finite horizon $T \geq 1$ we denote the benchmark cumulative log return by

$$B_T := \sum_{t=1}^T X_t. \quad (6)$$

Its variance and covariances with X_t and S_t can be expressed in terms of γ as

$$\text{Var}(B_T) = \sum_{u=1}^T \sum_{v=1}^T \gamma(u-v), \quad (7)$$

$$\text{Cov}(X_t, B_T) = \sum_{u=1}^T \gamma(t-u), \quad (8)$$

$$\text{Cov}(S_t, B_T) = \sum_{j=1}^k \text{Cov}(X_{t-j}, B_T) = \sum_{j=1}^k \sum_{u=1}^T \gamma(t-j-u). \quad (9)$$

3 Gaussian half-space identities

For any bivariate normal pair (X, S) with means (m_x, m_s) , variances (v_x, v_s) and covariance c_{sx} , the following identities hold:

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = m_x \Phi\left(\frac{m_s}{\sqrt{v_s}}\right) + \frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right), \quad (10)$$

$$\mathbb{E}[X \text{ sign}(S)] = 2 \mathbb{E}[X \mathbb{1}\{S > 0\}] - m_x. \quad (11)$$

These will be used repeatedly with $(X, S) = (X_t, S_t)$ and, in the conditional setting, with $(X, S) = (X_t, S_t)$ given $B_T = b$. A detailed derivation of these identities is provided in Appendix A.1.

4 Expected Log Return

4.1 One-Period Results

Using the Gaussian half-space identities with $(X, S) = (X_t, S_t)$ and the unconditional moments from Section 2, the expected log payoff is

$$\boxed{\mathbb{E}[Y_t] = \mathbb{E}[D_t X_t] = \mathbb{E}[X_t \text{ sign}(S_t)] = \mu \left(2\Phi\left(\frac{k\mu}{\sigma_S}\right) - 1 \right) + 2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right)} \quad (12)$$

where σ_S^2 and c_{sx} are determined by γ as above.

The unconditional expected payoff can be decomposed into two terms. ***Drift times average position*** and ***timing edge from the covariance***. The first term, $\mu(2\Phi(k\mu/\sigma_S) - 1)$, the unconditional drift of the underlying, μ , times the unconditional average position, $\mathbb{E}[D_t] = 2\Phi(k\mu/\sigma_S) - 1$. It thus represents the “directional exposure” of the strategy to the unconditional drift: if the signal is on average long ($\mu > 0$), this term is positive and the strategy implicitly participates in the benchmark drift.

The second term depends on c_{sx}/σ_S , i.e. the unconditional covariance between signal and next-period return, scaled by the unconditional volatility of the signal. This captures the extra edge from *timing*: being more likely to be long when the next return is high and short when it is low.

The Gaussian factor $2\phi(k\mu/\sigma_S)$ arises from the half-space integral and shrinks as the standardized signal mean $k\mu/\sigma_S$ moves far from zero, reflecting that the threshold at $S_t = 0$ cuts through a thinner part of the signal distribution when the rule is strongly biased long or short.

4.2 Multi-Period Results

For horizon $T \geq 1$, define the cumulative log payoff

$$\sum_{t=1}^T Y_t = \sum_{t=1}^T D_t X_t. \quad (13)$$

By stationarity of (D_t, X_t) ,

$$\mathbb{E}\left[\sum_{t=1}^T Y_t\right] = T \mathbb{E}[D_t X_t] = T \left(\mu \left(2\Phi\left(\frac{k\mu}{\sigma_S}\right) - 1 \right) + 2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right) \right) \quad (14)$$

with c_{sx} and σ_S^2 determined by the autocovariance function γ .

When the underlying asset has a positive drift, the trend strategy will thus also yield a positive return even when there is no positive contribution from the timing component.

5 Conditional Multi-Period Expected Returns

To analyze how the performance of a trend strategy depends upon the underlying, we will fix a realized benchmark return $B_T = b$, and study the performance of the trend strategy given some b . Under the stationary Gaussian model, the vector (X_t, S_t, B_T) is jointly normal, so conditioning on $B_T = b$ preserves normality of (X_t, S_t) with computable conditional moments. The Gaussian half-space identities from Section 3 then apply with conditional means, variances and covariances. A detailed derivation of the conditional parameters and the extension of the half-space identities to the conditional setting is provided in Appendix A.2.

Conditional parameters under a general Gaussian model

Using the autocovariance $\gamma(h)$, the conditional means and (co)variances of (X_t, S_t) given $B_T = b$ depend on t . We denote them by:

$$m_x^{(t)}(b) := \mathbb{E}[X_t \mid B_T = b] = \mu + \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)} (b - T\mu), \quad (15)$$

$$m_s^{(t)}(b) := \mathbb{E}[S_t \mid B_T = b] = k\mu + \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)} (b - T\mu), \quad (16)$$

$$v_x^{(t)}(b) := \text{Var}(X_t \mid B_T = b) = \sigma^2 - \frac{\text{Cov}(X_t, B_T)^2}{\text{Var}(B_T)}, \quad (17)$$

$$v_s^{(t)}(b) := \text{Var}(S_t \mid B_T = b) = \text{Var}(S_t) - \frac{\text{Cov}(S_t, B_T)^2}{\text{Var}(B_T)}, \quad (18)$$

$$c_{sx}^{(t)}(b) := \text{Cov}(S_t, X_t \mid B_T = b) = \text{Cov}(S_t, X_t) - \frac{\text{Cov}(S_t, B_T) \text{Cov}(X_t, B_T)}{\text{Var}(B_T)}. \quad (19)$$

Here all covariances involving X_t or S_t depend on t and are determined by $\gamma(h)$ via the formulas in Section 2. Note that while the conditional means depend on the realized value b , the conditional variances and covariance $(v_x^{(t)}(b), v_s^{(t)}(b), c_{sx}^{(t)}(b))$ are deterministic (constant in b) for Gaussian systems, though they vary with t .

Conditional expectations

The conditional multi-period log payoff is

$$\mathbb{E} \left[\sum_{t=1}^T Y_t \mid B_T = b \right] = \sum_{t=1}^T \left(m_x^{(t)}(b) \left[2\Phi \left(\frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \right) - 1 \right] + 2 \frac{c_{sx}^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \phi \left(\frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \right) \right) \quad (20)$$

obtained by applying the half-space identities to the conditional bivariate normal pair $(X_t, S_t \mid B_T = b)$ for each t . The justification that $(X_t, S_t \mid B_T = b)$ is bivariate normal with the stated parameters and that the Gaussian half-space identities extend to this conditional setting is provided in Appendix A.1.

The conditional multi-period payoff (20) is a direct analog to the unconditional payoff in (14) where the unconditional moments are replaced by their conditional counterparts. The drift component is given by the conditional drift of the underlying times the conditional average expected position. The timing component is given by the conditional alignment between the signal and return.

6 Hedge properties

This section records the key asymptotic features of the conditional multi-period log payoff under the general stationary Gaussian model. We distinguish between two regimes: a fixed horizon T

with an arbitrarily deep crash ($b \rightarrow -\infty$), and an infinite horizon $T \rightarrow \infty$ with a linear drawdown.

6.1 Fixed Horizon: Deep Crash ($b \rightarrow -\infty$)

Fix the horizon $T \geq 1$. We analyze the conditional expected payoff as the cumulative return $B_T = b$ tends to $-\infty$. Recall the decomposition:

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = \sum_{t=1}^T \left(m_x^{(t)}(b) [2\Phi(u_t(b)) - 1] + 2 \frac{c_{sx}^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \phi(u_t(b)) \right). \quad (21)$$

The asymptotic behavior of each term is governed by the standardized conditional signal mean

$$u_t(b) = \frac{m_s^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} = \frac{k\mu + \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)}(b - T\mu)}{\sqrt{\sigma_S^2 - \frac{\text{Cov}(S_t, B_T)^2}{\text{Var}(B_T)}}}, \quad (22)$$

where the key time-dependent covariance and benchmark variance are given by sums of autocovariances:

$$\text{Cov}(S_t, B_T) = \sum_{j=1}^k \sum_{u=1}^T \gamma(t-j-u), \quad \text{Var}(B_T) = \sum_{u=1}^T \sum_{v=1}^T \gamma(u-v).$$

Since $m_s^{(t)}(b)$ is linear in b , we distinguish three cases for each t based on the sign of the signal-benchmark covariance. For large $|b|$, the conditional means are dominated by their linear terms:

$$m_s^{(t)}(b) \sim \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)} b, \quad m_x^{(t)}(b) \sim \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)} b. \quad (23)$$

We introduce the index sets

$$\mathcal{T}_+ := \{t : \text{Cov}(S_t, B_T) > 0\}, \quad \mathcal{T}_- := \{t : \text{Cov}(S_t, B_T) < 0\}, \quad \mathcal{T}_0 := \{t : \text{Cov}(S_t, B_T) = 0\}.$$

1. **Positively correlated signal** ($t \in \mathcal{T}_+$): $\text{Cov}(S_t, B_T) > 0$. Here the coefficient $\frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)} > 0$, so as $b \rightarrow -\infty$, we have $m_s^{(t)}(b) \rightarrow -\infty$ and thus $u_t(b) \rightarrow -\infty$ linearly in b . By Mills' ratio, $\Phi(u_t(b))$ and $\phi(u_t(b))$ vanish exponentially fast (as e^{-cb^2}). The term $2\Phi(u_t(b)) - 1$ converges to -1 exponentially fast, and the timing term involving $\phi(u_t(b))$ vanishes. Hence the contribution is asymptotically $-m_x^{(t)}(b)$.
2. **Negatively correlated signal** ($t \in \mathcal{T}_-$): $\text{Cov}(S_t, B_T) < 0$. In this case the coefficient $\frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)} < 0$, so as $b \rightarrow -\infty$, we have $m_s^{(t)}(b) \rightarrow +\infty$ and thus $u_t(b) \rightarrow +\infty$. Then $2\Phi(u_t(b)) - 1 \rightarrow +1$ and $\phi(u_t(b)) \rightarrow 0$ exponentially fast, so the timing term again vanishes and the contribution is asymptotically $+m_x^{(t)}(b)$.
3. **Uncorrelated signal** ($t \in \mathcal{T}_0$): $\text{Cov}(S_t, B_T) = 0$. Here $m_s^{(t)}(b) = k\mu$ is constant, so $u_t(b) \equiv \frac{k\mu}{\sigma_S}$ is constant. The term $2\Phi\left(\frac{k\mu}{\sigma_S}\right) - 1$ is constant. The timing term is $2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right)$, which

vanishes exactly if the unconditional covariance $c_{sx} = 0$ (e.g. for i.i.d. returns with S_t built from strictly past returns). Even if non-zero, it is a constant $\mathcal{O}(1)$ term. Since the main term $m_x^{(t)}(b)$ grows linearly with b , the constant timing term becomes negligible asymptotically. Thus, the contribution is dominated by $m_x^{(t)}(b)(2\Phi(\frac{k\mu}{\sigma_S}) - 1)$.

General Asymptotic Formula. Summing these contributions, we obtain the general left-tail asymptotic

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] \sim -\sum_{t \in \mathcal{T}_+} m_x^{(t)}(b) + \sum_{t \in \mathcal{T}_-} m_x^{(t)}(b) + \sum_{t \in \mathcal{T}_0} m_x^{(t)}(b) \left(2\Phi\left(\frac{k\mu}{\sigma_S}\right) - 1\right), \quad b \rightarrow -\infty, \quad (24)$$

which is precise up to exponentially small error terms $\mathcal{O}(e^{-cb^2})$. The factor $2\Phi(\frac{k\mu}{\sigma_S}) - 1$ is precisely $\mathbb{E}[D_t]$, i.e. the unconditional average position. In the special case where there are no negatively correlated periods ($\mathcal{T}_- = \emptyset$), this reduces to the simpler form used below in the strictly positive correlation and i.i.d. examples.

Case 1: Strictly Positive Correlation (Perfect Hedge). If $\text{Cov}(S_t, B_T) > 0$ for all t (i.e., $\mathcal{T}_- = \mathcal{T}_0 = \emptyset$), then all periods contribute $-m_x^{(t)}(b)$ asymptotically. Since $\sum_{t=1}^T m_x^{(t)}(b) = \mathbb{E}[B_T \mid B_T = b] = b$, the total payoff becomes

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = -\sum_{t=1}^T m_x^{(t)}(b) + \mathcal{O}(e^{-cb^2}) = -b + \mathcal{O}(e^{-cb^2}). \quad (25)$$

Thus, strictly positive correlation across the entire horizon is a sufficient condition for the strategy to act as a perfect hedge (asymptotically) in a fixed-horizon crash.

Case 2: i.i.d. Benchmark. Consider the i.i.d. case. For $t = 1$, S_1 uses pre-sample data, so $\text{Cov}(S_1, B_T) = 0 \implies 1 \in \mathcal{T}_0$. For $t \geq 2$, S_t overlaps with B_T , so $\text{Cov}(S_t, B_T) > 0 \implies t \in \mathcal{T}_+$. Thus $\mathcal{T}_0 = \{1\}$. By the i.i.d. symmetry, each period contributes equally to the conditional mean: $m_x^{(t)}(b) = b/T$ for all t . Substituting into the general formula:

$$\text{Payoff} \sim -b + 2\frac{b}{T}\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) = b\left(\frac{2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right)}{T} - 1\right). \quad (26)$$

In our model, the signal is the sum of k returns: $S_t = \sum_{j=1}^k X_{t-j}$. For the i.i.d. benchmark, $\mathbb{E}[S_t] = k\mu$ and $\text{Var}(S_t) = k\sigma^2$. Thus the standardized signal mean simplifies to:

$$\frac{k\mu}{\sigma_S} = \frac{k\mu}{\sqrt{k}\sigma} = \frac{\mu}{\sigma}\sqrt{k}.$$

Since $0 \leq \Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) \leq 1$, the coefficient $\frac{2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right)}{T} - 1$ lies in $[-1, -1 + 2/T]$. For $T \geq 2$, it is strictly negative, meaning the payoff is positive (since $b < 0$).

For the special case $T = 1$, the coefficient becomes $2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) - 1$. If the asset has positive drift ($\mu > 0$), then $\frac{\mu}{\sigma}\sqrt{k} > 0$ and $\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) > 1/2$, so the coefficient is positive and the payoff $\sim b(2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) - 1)$ is negative (a loss). Intuitively, with $T = 1$, the single return X_1 constitutes the entire crash. Since the signal S_1 (based on past data) is uncorrelated with X_1 , the strategy cannot anticipate the move. Instead, it relies on its unconditional bias: if $\mu > 0$, the strategy is usually long ($2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) - 1 > 0$). When the crash $X_1 = b$ hits, this long bias results in a loss proportional to the crash size. This highlights that the "perfect hedge" property requires at least one period of look-ahead or correlation ($T \geq 2$) to adapt.

Trade-off between Expected Return and Hedging. The deep-crash slope $\frac{2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right)}{T} - 1$ reveals a fundamental trade-off between unconditional expected returns and hedging effectiveness. With i.i.d. returns, the unconditional expected trend payoff is solely driven by the drift component $\mu\left(2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right) - 1\right)$, which is positive when $\mu > 0$ and increasing in k . However, the same quantity $\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right)$ that increases this drift component also increases the deep-crash slope, weakening the hedge. This tension reflects the intuitive tradeoff when choosing the lookback period: a longer lookback gives a higher expected return but lower hedge effectiveness.

6.2 Infinite Horizon: Linear Drawdown ($T \rightarrow \infty$)

Consider the limit $T \rightarrow \infty$ where the benchmark realizes a linear drawdown $B_T = Td$ with $d < 0$. We assume the process (X_t) has short memory (specifically, absolutely summable autocovariance $\sum |\gamma(h)| < \infty$). This ensures that the benchmark variance $\text{Var}(B_T)$ grows linearly in T , while covariances like $\text{Cov}(X_t, B_T)$ remain bounded. Consequently, the conditional variance $\text{Var}(X_t \mid B_T = b)$ converges to the unconditional variance σ^2 as $T \rightarrow \infty$.

Convergence of Conditional Moments. Using the conditional mean formula derived in Section 5:

$$m_x^{(t)}(b) = \mu + \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)}(b - T\mu).$$

Substituting $b = Td$, the shift term becomes:

$$\frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)} \cdot (T(d - \mu)).$$

For a short-memory process, $\text{Var}(B_T) \sim V \cdot T$ and $\text{Cov}(X_t, B_T) \sim V$ (where V is the long-run variance). Thus, the ratio $\frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)} \sim \frac{V}{VT} = \frac{1}{T}$. The total shift is approximately $\frac{1}{T} \cdot T(d - \mu) = d - \mu$. Consequently, the conditional mean converges to:

$$\lim_{T \rightarrow \infty} m_x^{(t)}(Td) = d.$$

By linearity, the signal mean shifts by k times this amount:

$$\lim_{T \rightarrow \infty} m_s^{(t)}(Td) = kd.$$

The conditional variances converge to their unconditional values. Recall that $v_x(b) = \sigma^2 - \frac{\text{Cov}(X_t, B_T)^2}{\text{Var}(B_T)}$. For a short-memory process, the covariance $\text{Cov}(X_t, B_T)$ is bounded (converging to V), while the benchmark variance $\text{Var}(B_T)$ grows linearly with T . Thus, the variance reduction term scales as $\frac{V^2}{VT} = \mathcal{O}(1/T)$, which vanishes as $T \rightarrow \infty$. Intuitively, knowing the average over an infinite horizon provides negligible information about any specific local fluctuation.

Limiting Per-Period Payoff. Recall the general conditional payoff formula from (20):

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = \sum_{t=1}^T \left(m_x^{(t)}(b) [2\Phi(u_t(b)) - 1] + 2 \frac{c_{sx}^{(t)}(b)}{\sqrt{v_s^{(t)}(b)}} \phi(u_t(b)) \right). \quad (27)$$

As $T \rightarrow \infty$ with $b = Td$, we substitute the limiting moments derived above into this summand. For any interior t , the conditional mean $m_x^{(t)}(Td) \rightarrow d$ and the standardized signal mean

$$u_t(Td) = \frac{m_s^{(t)}(Td)}{\sqrt{v_s^{(t)}(Td)}} \rightarrow \frac{kd}{\sigma_S}.$$

The coefficient $c_{sx}^{(t)}(Td)/\sqrt{v_s^{(t)}(Td)} \rightarrow \frac{c_{sx}}{\sigma_S}$ converges to the unconditional signal-return alignment, where $c_{sx} = \text{Cov}(S_t, X_t)$ is the unconditional covariance from Section 2 (independent of t by stationarity). Thus, each term in the sum converges to the per-period limit

$$g(d) := \lim_{T \rightarrow \infty} \left(m_x^{(t)}(Td) [2\Phi(u_t(Td)) - 1] + 2 \frac{c_{sx}^{(t)}(Td)}{\sqrt{v_s^{(t)}(Td)}} \phi(u_t(Td)) \right), \quad (28)$$

which evaluates to

$$g(d) = \underbrace{d \left[2\Phi\left(\frac{kd}{\sigma_S}\right) - 1 \right]}_{\text{Trend Component}} + \underbrace{2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{kd}{\sigma_S}\right)}_{\text{Timing Component}}. \quad (29)$$

Since there are T such terms (and boundary effects are negligible), the total expected payoff becomes:

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = Td\right] = T \cdot g(d) + \mathcal{O}(1). \quad (30)$$

Intuitively, as $T \rightarrow \infty$, the cumulative systematic component $T \cdot g(d)$ grows linearly with the horizon, while local fluctuations around individual returns average out. Thus the deterministic drift term dominates the stochastic noise, and the per-period limit $g(d)$ fully characterizes the strategy's long-run performance.

Sign Analysis and Threshold Property. The function $g(d)$ determines the long-run profitability of the strategy during a sustained drawdown.

- **Trend Component:** $d[2\Phi(kd/\sigma_S) - 1]$. Since $d < 0$, we have $kd/\sigma_S < 0$ (negative signal mean). The position bias $2\Phi(kd/\sigma_S) - 1$ is then negative (short bias). Because $d < 0$, the product is positive. This term represents the profit from the strategy correctly identifying the downward trend and maintaining a persistent short position.
- **Timing Component:** $2\frac{c_{sx}}{\sigma_S}\phi(kd/\sigma_S)$. This term captures the profit from local correlations between the signal and the subsequent return. Its sign depends on c_{sx} .
 - If $c_{sx} > 0$ (unconditional momentum), this term is positive, reinforcing the profit.
 - If $c_{sx} < 0$ (mean reversion), this term is negative, acting as a drag on performance.

Threshold Property: If $c_{sx} \geq 0$, then $g(d) > 0$. The strategy is always profitable. If $c_{sx} < 0$, there is a competition between the linear Trend Component (profit) and the Timing Component (loss). However, as $d \rightarrow -\infty$, the Trend Component grows linearly ($\sim |d|$), while the Timing Component decays exponentially ($\phi(kd/\sigma_S) \sim e^{-d^2}$). Therefore, there exists a critical threshold d^* such that for all $d < d^*$, $g(d) > 0$. This implies that *any* stationary Gaussian process with short memory allows for profitable trend following during sufficiently steep linear drawdowns, even if the process is locally mean-reverting. The "trend" eventually overwhelms the "noise".

Lookback and Hedging: Competing Effects. The effect of the lookback parameter k on hedging in the linear drawdown regime is governed by two competing forces in the standardized signal mean kd/σ_S . The numerator kd captures the trend: a longer lookback accumulates more of the drawdown $d < 0$, strengthening the short signal. The denominator σ_S represents signal noise: a longer lookback also increases signal volatility. The net effect on $|kd/\sigma_S|$ depends on the correlation structure. For i.i.d. returns, $\sigma_S = \sqrt{k}\sigma$, so $|kd/\sigma_S| = \sqrt{k}|d|/\sigma$ grows as $k^{1/2}$. For positively correlated (momentum) returns, σ_S grows faster than $\sqrt{k}$ due to positive cross-terms, so $|kd/\sigma_S|$ grows more slowly—or may not grow at all for highly persistent processes. For negatively correlated (mean-reverting) returns, σ_S grows slower than $\sqrt{k}$ due to negative cross-terms, so $|kd/\sigma_S|$ grows faster. For mean-reverting processes, a higher lookback period can thus improve the hedging properties by better capturing the trend down with a limited cost in terms of signal noise. For highly persistent processes, a shorter lookback period on the contrary allows a faster reaction to a downtrend.

7 ARMA(p,q) Specialization

The preceding sections require only that (X_t) be a stationary Gaussian process with autocovariance $\gamma(h)$ and that γ be short-memory (e.g. absolutely summable) for the linear-drawdown asymptotics. A broad and important class of processes satisfying these conditions is given by stationary Gaussian

ARMA(p,q) models, which have been widely used in financial econometrics (Campbell et al., 1997). The evidence that stock returns exhibit short-term positive autocorrelation (Lo and MacKinlay, 1988) provides empirical motivation for studying trend strategies under such processes.

Assume now that (X_t) follows a Gaussian ARMA(p,q) process

$$X_t - \mu = \sum_{j=1}^p \phi_j (X_{t-j} - \mu) + \sum_{\ell=0}^q \theta_\ell \varepsilon_{t-\ell}, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2), \quad (31)$$

with the usual stationarity and invertibility conditions on the AR and MA polynomials. Under these assumptions:

- (X_t) is stationary and Gaussian with mean μ and variance σ^2 ,
- the autocovariance function $\gamma(h) = \text{Cov}(X_t, X_{t+h})$ is determined uniquely by the ARMA parameters and is absolutely summable (short memory),
- hence all the general results derived above (one-period formulas, conditional multi-period expression, fixed- T left-tail behavior for $T \geq 2$, and the linear-drawdown limits) apply directly with this ARMA autocovariance $\gamma(h)$.

In particular, the ARMA(p,q) specialization is “admissible” in the sense that it satisfies all structural requirements of the Gaussian framework: joint normality of (X_t, S_t, B_T) , stationarity, and the short-memory condition needed for the asymptotic analysis in Section 6. Plugging the ARMA autocovariance function $\gamma(h)$ into the formulas in Sections 2–6 yields explicit expressions for $\text{Var}(S_t)$, $\text{Cov}(S_t, X_t)$, $\text{Var}(B_T)$, $\text{Cov}(X_t, B_T)$ and $\text{Cov}(S_t, B_T)$ in terms of the ARMA parameters, and hence explicit versions of all central results for the ARMA(p,q) case.

Example: Fixed-Horizon Deep Crash for a Gaussian MA(1)

As a concrete illustration of the fixed-horizon deep-crash asymptotics in Section 6.1, consider a Gaussian MA(1) benchmark

$$X_t - \mu = \varepsilon_t + \theta \varepsilon_{t-1}, \quad \varepsilon_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\varepsilon^2), \quad (32)$$

with the usual invertibility condition $|\theta| < 1$. The autocovariance function is

$$\gamma(0) = (1 + \theta^2)\sigma_\varepsilon^2 =: \gamma_0 > 0, \quad \gamma(1) = \gamma(-1) = \theta\sigma_\varepsilon^2 =: \gamma_1, \quad \gamma(h) = 0 \text{ for } |h| \geq 2.$$

For the T -period benchmark log return $B_T = \sum_{t=1}^T X_t$ we obtain

$$\text{Var}(B_T) = T\gamma_0 + 2(T-1)\gamma_1.$$

For $T \geq 2$, the covariances between X_t and B_T take the simple form

$$\text{Cov}(X_1, B_T) = \text{Cov}(X_T, B_T) = \gamma_0 + \gamma_1, \quad \text{Cov}(X_t, B_T) = \gamma_0 + 2\gamma_1, \quad 2 \leq t \leq T-1.$$

Consequently, the conditional means admit the representation

$$m_x^{(t)}(b) := \mathbb{E}[X_t | B_T = b] = \mu + \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)}(b - T\mu) = \alpha_t b + \mathcal{O}(1), \quad \sum_{t=1}^T \alpha_t = 1,$$

with

$$\alpha_1 = \alpha_T = \frac{\gamma_0 + \gamma_1}{T\gamma_0 + 2(T-1)\gamma_1}, \quad \alpha_t = \frac{\gamma_0 + 2\gamma_1}{T\gamma_0 + 2(T-1)\gamma_1}, \quad 2 \leq t \leq T-1.$$

Before determining the sign of the conditional payoff, we record the covariance between the signal and the benchmark return. Since $\gamma(h) = 0$ for $|h| \geq 2$, only the first lag contributes to the correlation between past and future. For $t = 1$, S_1 involves $X_0, \dots, X_{1-k}$. Only X_0 correlates with B_T (via X_1), so

$$\text{Cov}(S_1, B_T) = \gamma_1. \quad (33)$$

For $t \geq 2$, S_t involves X_{t-1} , which is part of the benchmark sum B_T . Specifically, we can verify that $\text{Cov}(S_t, B_T) > 0$ for all $t = 2, \dots, T$ by inspecting the contributions of the individual returns X_v in the signal sum $S_t = \sum_{v=t-k}^{t-1} X_v$. The covariance of any single return X_v with B_T is:

- For interior terms $2 \leq v \leq T-1$: $\text{Cov}(X_v, B_T) = \gamma_0 + 2\gamma_1 = (1 + \theta)^2 \sigma_\varepsilon^2 > 0$.
- For boundary terms $v = 1$ or $v = T$: $\text{Cov}(X_v, B_T) = \gamma_0 + \gamma_1 = (1 + \theta + \theta^2) \sigma_\varepsilon^2 > 0$. (Note that $1 + \theta + \theta^2 > 0$ for all real θ).
- For the immediate pre-sample term $v = 0$: $\text{Cov}(X_0, B_T) = \gamma_1 = \theta \sigma_\varepsilon^2$.
- For older terms $v < 0$: $\text{Cov}(X_v, B_T) = 0$.

For $t \geq 2$, the signal always contains X_{t-1} with $t-1 \geq 1$. If the signal is short or t is large such that all components X_v have $v \geq 1$, then all covariance contributions are strictly positive. If the signal extends into the pre-sample period ($t-k \leq 0$), it includes X_0 (which contributes γ_1) but must also include X_1 (which contributes $\gamma_0 + \gamma_1$). The sum of these two terms is $\gamma_0 + 2\gamma_1 = (1 + \theta)^2 \sigma_\varepsilon^2 > 0$. Thus, the positive correlation from the overlapping periods always dominates any negative correlation from the pre-sample period, ensuring:

$$\text{Cov}(S_t, B_T) > 0, \quad t = 2, \dots, T. \quad (34)$$

This sign pattern—determined entirely by γ_1 at $t = 1$ and positive thereafter—drives the results in the three regimes below.

In the deep-crash regime $b \rightarrow -\infty$ with fixed T , the general Gaussian analysis in Section 6.1 shows that the sign of the conditional payoff is governed by the signs of the signal–benchmark covariances $\text{Cov}(S_t, B_T)$. For the MA(1) process this leads to three qualitatively distinct regimes:

- **Positive MA(1) correlation** ($\theta > 0$). All nonzero autocovariances $\gamma(h)$ are positive, and both S_t and B_T are nonnegative linear combinations of the same returns $\{X_u\}$. Hence

$$\text{Cov}(S_t, B_T) > 0 \quad \text{for all } t = 1, \dots, T.$$

This is exactly the “strictly positive correlation” case in Section 6.1. The deep-crash payoff simplifies to

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] = -b + \mathcal{O}(e^{-cb^2}), \quad b \rightarrow -\infty,$$

for some $c > 0$ depending on the model. Thus, for any fixed $T \geq 1$ and any lookback $k \geq 1$, a Gaussian MA(1) benchmark with $\theta > 0$ yields an asymptotically *perfect hedge* in a finite-horizon crash.

- **i.i.d. benchmark** ($\theta = 0$). When $\theta = 0$, the MA(1) reduces to the i.i.d. Gaussian case already analyzed in Section 6.1 (Case 2). One has $\mathcal{T}_0 = \{1\}$ and, by symmetry, $m_x^{(t)}(b) = b/T$, which leads to

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] \sim b \left(\frac{2\Phi\left(\frac{\mu}{\sigma}\sqrt{k}\right)}{T} - 1 \right), \quad b \rightarrow -\infty.$$

For $T \geq 2$ this coefficient is strictly negative, hence the conditional payoff is asymptotically positive in a deep crash; for $T = 1$ the sign depends on the unconditional drift through μ (equivalently through $\frac{\mu}{\sigma}\sqrt{k}$).

- **Locally mean-reverting MA(1)** ($\theta < 0$). For an invertible MA(1) with $\theta < 0$, the process is locally mean-reverting. In this case the signal–benchmark covariance is negative at $t = 1$ and positive for $t \geq 2$, so that

$$\mathcal{T}_- = \{1\}, \quad \mathcal{T}_+ = \{2, \dots, T\}, \quad \mathcal{T}_0 = \emptyset,$$

where $\mathcal{T}_\pm$ denote the sets of times with $\text{Cov}(S_t, B_T) \gtrless 0$. The general deep-crash expansion in Section 6.1 specializes in this case to

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] \sim -\sum_{t \in \mathcal{T}_+} m_x^{(t)}(b) + \sum_{t \in \mathcal{T}_-} m_x^{(t)}(b) = -\sum_{t=2}^T m_x^{(t)}(b) + m_x^{(1)}(b).$$

Using $\sum_{t=1}^T m_x^{(t)}(b) = b$ (since $\sum_{t=1}^T X_t = B_T$), we have $\sum_{t=2}^T m_x^{(t)}(b) = b - m_x^{(1)}(b)$ and hence

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] \sim -(b - m_x^{(1)}(b)) + m_x^{(1)}(b) = -b + 2m_x^{(1)}(b).$$

Since each conditional mean is linear in b of the form $m_x^{(t)}(b) = \alpha_t b + \mathcal{O}(1)$ with coefficients

$\alpha_t = \text{Cov}(X_t, B_T) / \text{Var}(B_T)$ satisfying $\sum_{t=1}^T \alpha_t = 1$, we obtain $m_x^{(1)}(b) \sim \alpha_1 b$ as $b \rightarrow -\infty$ and thus a linear asymptotic of the form

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] \sim (2\alpha_1 - 1)b, \quad b \rightarrow -\infty,$$

where $2\alpha_1 - 1$ is the leading-order coefficient (slope in b) of the conditional payoff, in direct analogy with the i.i.d. slope $\frac{2\Phi(\frac{\mu}{\sigma}\sqrt{k})}{T} - 1$. Substituting the explicit MA(1) weights yields

$$2\alpha_1 - 1 = -\frac{(T-2)(\gamma_0 + 2\gamma_1)}{T\gamma_0 + 2(T-1)\gamma_1} = -\frac{(T-2)(1+\theta)^2\sigma_\varepsilon^2}{T(1+\theta^2)\sigma_\varepsilon^2 + 2(T-1)\theta\sigma_\varepsilon^2}. \quad (35)$$

For any invertible MA(1) with $\theta > -1$ the denominator is positive and $(1+\theta)^2\sigma_\varepsilon^2 \geq 0$, so:

- If $T \geq 3$ and $\theta \neq -1$, then $2\alpha_1 - 1 < 0$ and

$$\mathbb{E}\left[\sum_{t=1}^T Y_t \mid B_T = b\right] \sim (2\alpha_1 - 1)b > 0, \quad b \rightarrow -\infty.$$

Thus even for a locally mean-reverting MA(1) benchmark, any crash lasting at least three periods is asymptotically profitable for the trend rule.

- **Special case** $T = 2$. Here $\text{Var}(B_2) = 2\gamma_0 + 2\gamma_1 = 2(\gamma_0 + \gamma_1)$ and $\text{Cov}(X_1, B_2) = \gamma_0 + \gamma_1$, so

$$\alpha_1 = \frac{\text{Cov}(X_1, B_2)}{\text{Var}(B_2)} = \frac{\gamma_0 + \gamma_1}{2(\gamma_0 + \gamma_1)} = \frac{1}{2},$$

independently of θ (and of the signal length k). Consequently

$$2\alpha_1 - 1 = 0.$$

In this special case, symmetry implies that $\alpha_1 = \alpha_2 = 1/2$, so the conditional means are $m_x^{(1)}(b) = m_x^{(2)}(b) = b/2$ exactly (the constant drift μ cancels). Since the asymptotic payoff is $-m_x^{(2)}(b) + m_x^{(1)}(b) = -b/2 + b/2 = 0$, the cancellation is exact. The residual terms decay exponentially, so

$$\lim_{b \rightarrow -\infty} \mathbb{E}\left[\sum_{t=1}^2 Y_t \mid B_2 = b\right] = 0.$$

Thus, for $T = 2$ and $\theta < 0$, the strategy neither profits nor loses significantly in an arbitrarily deep crash; the expected payoff vanishes.

- If $T = 1$, the strategy cannot anticipate the single crash period and the conditional payoff is asymptotically negative.

This MA(1) example shows that the general deep-crash theory in Section 6.1 can be made fully

explicit in a simple parametric setting: for positively correlated benchmarks ($\theta \geq 0$) the trend rule provides an asymptotically perfect hedge, while even for locally mean-reverting benchmarks ($\theta < 0$) the rule becomes profitable once the crash is sufficiently long.

Numerical Illustrations

Figures 1–3 illustrate these theoretical findings. Figure 1 displays the unconditional expected log payoff as a function of the MA parameter θ and lookback k . As expected, positive serial correlation ($\theta > 0$) yields high payoffs (momentum), while strong mean reversion ($\theta < 0$) can lead to losses unless the lookback k is tuned to capture longer-term trends. Figure 2 shows the conditional payoff profile, highlighting the convexity (“smile”) characteristic of trend strategies in favorable regimes. Finally, Figure 3 confirms the asymptotic analysis: for deep crashes ($b \ll 0$), the simulated conditional payoff converges precisely to the theoretical linear limit $(2\alpha_1 - 1)b$, demonstrating the robustness of the derived asymptotic slopes.

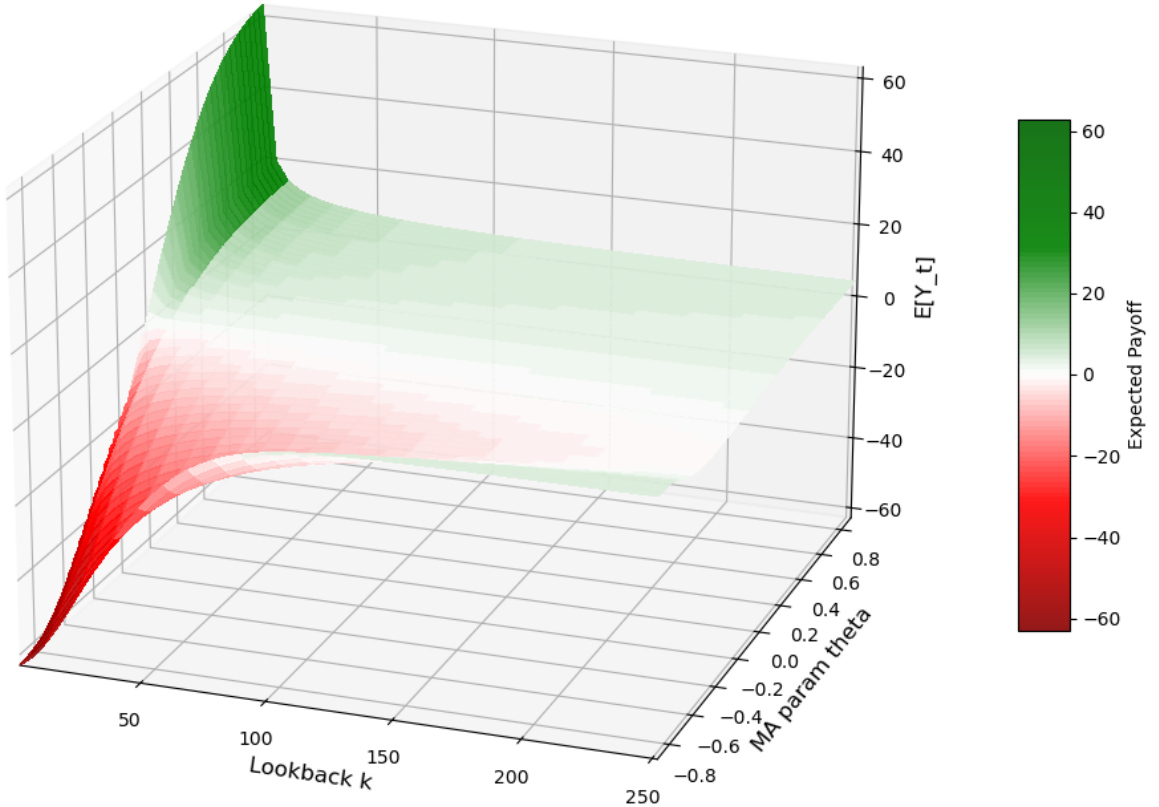


Figure 1: Annual Unconditional Payoff Illustration with underlying MA(1) Process (Sharpe 0.5)

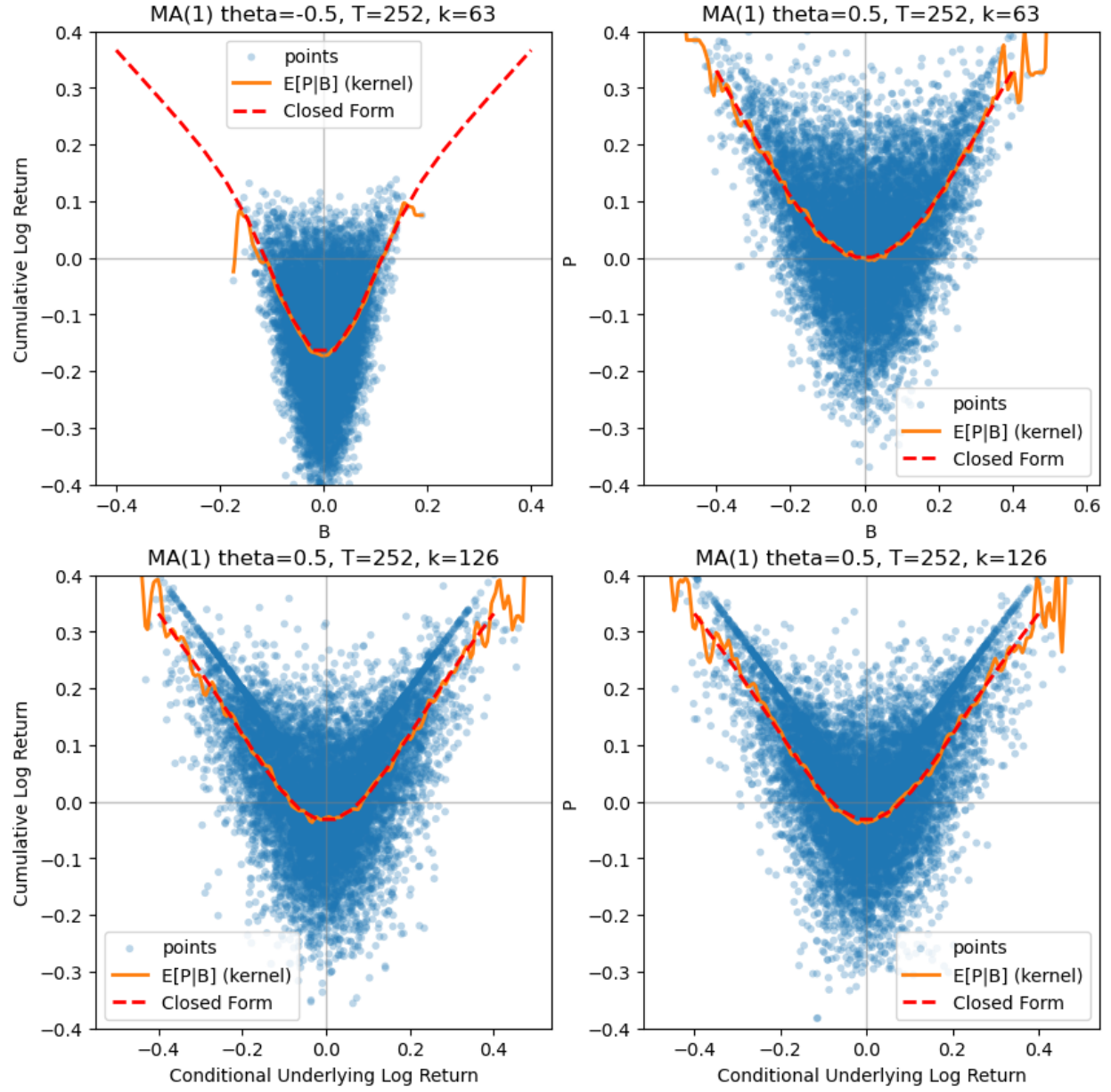


Figure 2: Annual Conditional Payoff Illustration with underlying MA(1) Process (Sharpe 0.5)

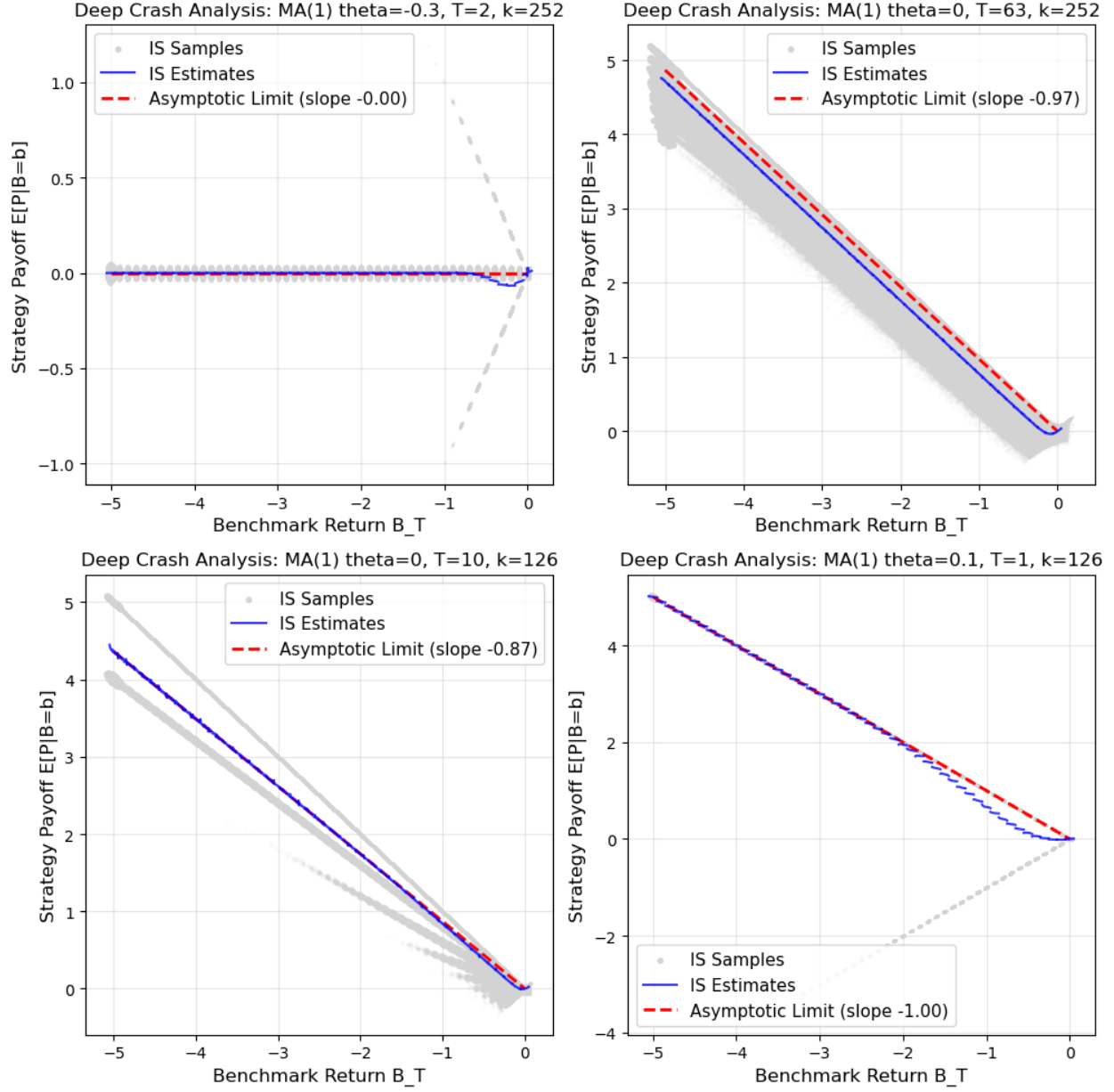


Figure 3: Fixed Horizon Deep Crash limit illustrations with underlying MA(1) Process

8 Empirical Analysis

This section tests the key theoretical predictions using a multi-asset dataset of daily returns. Our empirical analysis complements prior studies documenting trend-following performance across asset classes and time periods (Moskowitz et al., 2012; Hurst et al., 2017; Lempérière et al., 2014).

8.1 Data and Implementation

The dataset comprises daily log excess returns for 103 liquid futures and forward contracts spanning four major asset classes: Equity Indices (e.g., S&P 500, Nikkei 225), Government Bonds (e.g., US Treasuries, German Bunds), Commodities (e.g., Crude Oil, Gold, Copper), and Credit. The sample period covers December 1999 to August 2025, providing approximately 6,700 daily observations per asset. To ensure comparability across diverse instrument types, each log return series is ex-ante scaled to a constant 10% annualized volatility target (daily standard deviation $\sigma \approx 0.63\%$).

The trend strategy uses signal $S_t = \sum_{j=1}^k \log(1 + R_{t-j})$ with position $D_t = \text{sign}(S_{t-1})$, where the one-day lag reflects realistic execution constraints.

8.2 Statistical Significance of Timing

To rigorously test the contribution of the timing component, we utilize the theoretical decomposition derived in Eq. (12). By the identity $\mathbb{E}[DX] = \mathbb{E}[D] \mathbb{E}[X] + \text{Cov}(D, X)$, the total expected payoff decomposes as

$$\underbrace{\mathbb{E}[D_t X_t]}_{\text{Total Return}} = \underbrace{\mathbb{E}[D_t] \mathbb{E}[X_t]}_{\text{Static Drift Component}} + \underbrace{\text{Cov}(D_t, X_t)}_{\text{Timing Component}}, \quad (36)$$

where the first term is the return from simply holding the average position $\mathbb{E}[D_t]$, and the second term captures the additional return from aligning the position with future returns. Under the Gaussian model of Section 2, this timing component equals the second term in Eq. (12), i.e. $2 \frac{c_{sx}}{\sigma_S} \phi\left(\frac{k\mu}{\sigma_S}\right)$; empirically, no distributional assumption is needed.

We test $H_0: \text{Cov}(D_t, X_t) = 0$ as follows. For each asset i and lookback k , define

$$\tau_{i,t}(k) := (D_{i,t}(k) - \bar{D}_i(k)) X_{i,t}, \quad \bar{D}_i(k) := \frac{1}{T} \sum_{t=1}^T D_{i,t}(k), \quad (37)$$

where $D_{i,t}(k) = \text{sign}(S_{i,t}(k))$ is the trend position based on lookback k . The sample mean $\frac{1}{T} \sum_t \tau_{i,t}(k)$ is the sample covariance $\widehat{\text{Cov}}(D_{i,t}(k), X_{i,t})$, so testing whether this mean is zero is a direct test of zero timing. Concretely, we regress

$$\tau_{i,t}(k) = \alpha_i(k) + \varepsilon_{i,t} \quad (38)$$

and test $H_0: \alpha_i(k) = 0$ using Newey-West HAC standard errors with k lags to account for the overlapping nature of the trend signal. The intercept $\alpha_i(k)$ is the timing component for asset i in daily return units.

For the aggregate, we follow the [Fama and MacBeth \(1973\)](#) methodology: at each date t , we compute the cross-sectional average

$$\bar{\tau}_t(k) := \frac{1}{N_t} \sum_{i=1}^{N_t} \tau_{i,t}(k), \quad (39)$$

where N_t is the number of assets with available data at date t , and test $H_0 : \mathbb{E}[\bar{\tau}_t(k)] = 0$ via Newey-West with k lags. This approach naturally accounts for cross-sectional dependence among assets, since at each date the correlated assets contribute to a single cross-sectional observation.

Table 1 reports the cross-sectional mean of the underlying asset return, the total trend strategy return, and its decomposition into drift ($\bar{D}_i(k) \cdot \bar{X}_i$) and timing components, along with statistical significance measures. The timing component is positive and economically meaningful across all lookback horizons, with the strongest statistical evidence ($t = 2.1$) at the one-year horizon ($k = 252$).

Table 1: Statistical Significance of Timing Component

Lookback (k)	Annualized Averages				FM t-stat	% Sig. Individ.
	Underlying	Trend	Drift	Timing		
21 Days	2.8%	2.0%	0.4%	1.5%	1.8	14%
63 Days	2.8%	1.8%	0.6%	1.2%	1.2	11%
126 Days	2.9%	1.5%	0.7%	0.8%	0.8	4%
252 Days	2.9%	2.8%	0.9%	1.8%	2.1	20%

Table 1 shows that the timing component explains a large part of the total trend strategy return, particularly at shorter horizons. For instance, with a 21-day lookback, timing contributes 1.5% of the 2.0% total return, implying that roughly 75% of the strategy’s performance derives from timing rather than passive exposure to the underlying drift. The drift component increases monotonically with the lookback horizon (from 0.4% at $k = 21$ to 0.9% at $k = 252$), exactly as the theory predicts: longer lookback windows produce a stronger directional bias $\bar{D}_i(k)$, increasing the strategy’s passive exposure to the underlying drift. The Fama-MacBeth t-statistics, which account for cross-asset correlation, are substantially lower than naive cross-sectional t-tests would imply, reflecting the considerable co-movement among the 103 assets. The one-year lookback ($k = 252$) yields the only timing component that is statistically significant at the 5% level ($t = 2.1$), while the one-month lookback is marginally significant ($t = 1.8$). The low individual significance rates (4–20%) relative to the positive timing means underscore that trend signals are noisy at the single-asset level. Finally, note that even though timing contributes positively, the total trend return is lower than the underlying.

8.3 Conditional Payoff Asymmetry

To rigorously test the non-linear payoff profile predicted by the theory, we employ a piecewise linear regression that distinguishes between up and down market regimes. We examine how the choice of lookback window k interacts with the evaluation horizon T . For each pair (k, T) , we follow the [Fama and MacBeth \(1973\)](#) methodology: at each date t , we run a cross-sectional regression across all assets

$$R_{i,t,T}^{\text{trend}}(k) = \alpha_t + \beta_{\text{down},t} \min(R_{i,t,T}^{\text{bench}}, 0) + \beta_{\text{up},t} \max(R_{i,t,T}^{\text{bench}}, 0) + \epsilon_{i,t}, \quad (40)$$

where $R_{i,t,T}^{\text{trend}}(k)$ is the overlapping T -day return of the trend strategy with lookback k for asset i , and $R_{i,t,T}^{\text{bench}}$ is the corresponding benchmark return. This produces a time series of coefficient estimates $\{\alpha_t, \beta_{\text{down},t}, \beta_{\text{up},t}\}$, whose time-series means are reported with Newey-West standard errors using $T + k$ lags to account for the overlap-induced autocorrelation. The Fama-MacBeth procedure naturally accounts for cross-sectional dependence among assets, as each date’s cross-sectional regression yields a single set of coefficients regardless of how correlated the assets are. As a further robustness check, we estimate the regressions on an equal-weighted portfolio of all assets (with Newey-West standard errors); the portfolio-level results (not reported) confirm the same qualitative patterns.

Table 2 reports the Fama-MacBeth mean coefficients $\beta_{\text{down}}/\beta_{\text{up}}$. The diagonal and near-diagonal elements ($T \approx k$) show the strongest asymmetry, with coefficients approaching unity (e.g., $-0.76/1.06$ for $T = k = 126$). This confirms that trend strategies act as effective straddles at their natural time horizon. The off-diagonal elements reveal the robustness of the hedging property. A short-term trend signal ($k = 21$) provides significant downside protection even against long-term crashes ($T = 252, \beta_{\text{down}} = -0.47$), whereas long-term signals are too slow to hedge sudden, short-term crashes ($k = 252, T = 21, \beta_{\text{down}} = 0.04$, not significant). This aligns with the theoretical insight that faster signals reduce lag but increase noise, creating a trade-off that depends on the crash duration.

Table 2: Conditional Asymmetry Matrix ($\beta_{\text{down}}/\beta_{\text{up}}$)

Horizon (T)	Lookback (k)			
	21 Days	63 Days	126 Days	252 Days
21 Days	-0.61** / 0.81**	-0.18** / 0.52**	-0.02 / 0.44**	0.04 / 0.46**
63 Days	-0.58** / 0.72**	-0.66** / 0.93**	-0.24** / 0.68**	-0.09 / 0.62**
126 Days	-0.60** / 0.57**	-0.80** / 0.92**	-0.76** / 1.06**	-0.26 / 0.82**
252 Days	-0.47** / 0.49**	-0.69** / 0.76**	-0.96** / 1.00**	-0.63** / 1.06**

** indicates Fama-MacBeth t-statistic significant at the 1% level.

Table 3 reports the annualized Fama-MacBeth regression intercept α , representing the “cost of convexity” or the option premium paid by the strategy. The cost is naturally highest for short horizons ($T = k = 21$, -17.4%), reflecting the high turnover and whipsaw losses associated with fast signals. For long evaluation horizons ($T = 252$), the absolute cost appears low across all lookbacks (approx. $2.5\text{--}6.4\%$). However, this lower absolute cost for short lookbacks ($k = 21$) must be weighed against their significantly reduced participation ($\beta_{\text{down}} = -0.47$ versus -0.63 for $k = 252$ in Table 2). Effectively, short-term signals provide a “smaller” option at long horizons—paying less premium but offering drastically less protection. This aligns with Table 1, where the $k = 21$ strategy generated the lowest total return, illustrating that while fast signals reduce lag, they are less efficient at capturing the full magnitude of long-term trends compared to longer lookbacks.

Table 3: Annualized Alpha (Cost of Convexity)

Horizon (T)	Lookback (k)			
	21 Days	63 Days	126 Days	252 Days
21 Days	-17.4%**	-8.8%**	-5.8%**	-3.7%**
63 Days	-9.2%**	-11.0%**	-7.0%**	-3.9%**
126 Days	-5.0%**	-8.0%**	-8.7%**	-4.5%**
252 Days	-2.5%**	-4.2%**	-6.4%**	-5.2%**

** indicates Fama-MacBeth t-statistic significant at the 1% level.

8.4 Drawdown Analysis

To analyze strategy performance during market stress, we conduct an event study of drawdown episodes. For each asset, we identify all peak-to-trough drawdowns exceeding 5% in depth using the underlying asset’s cumulative return. We then compute the cumulative trend return for different lookback windows (k) during these specific episodes.

Table 4 presents the average trend return across drawdown depth buckets for varying lookbacks. The results reveal a striking trade-off between signal speed and crash capture:

- **Shallow Corrections (5-10%):** Fast signals ($k = 21$) significantly outperform slow signals, losing only -1.1% compared to -5.5% for $k = 252$. Fast signals adapt quickly to the mean-reverting nature of minor corrections, whereas slow signals remain positioned wrongly throughout the episode.
- **Deep Crashes (>40%):** The relationship reverses. Slow signals ($k = 252$) generate the highest profits ($+31.1\%$), outperforming fast signals ($+17.3\%$). In extended crashes, slow signals effectively filter out the volatile counter-trend rallies that frequently whipsaw faster strategies, allowing them to capture the full magnitude of the decline.
- **Medium Drawdowns (20-40%):** Fast signals ($k = 21, 63$) provide superior protection ($+10.1\%, +6.1\%$), while slow signals ($k = 252$) are still too sluggish to react (-0.5%).

Table 4: Average Trend Return during Drawdown Episodes (Depth vs. Lookback)

Drawdown Depth	Lookback (k)			
	21 Days	63 Days	126 Days	252 Days
5-10%	-1.1%	-3.6%	-5.0%	-5.5%
10-20%	1.2%	-1.3%	-4.7%	-7.6%
20-40%	10.1%	6.1%	1.4%	-0.5%
>40%	17.3%	17.9%	23.7%	31.1%

This empirical evidence validates the theoretical threshold property while adding a crucial nuance: the optimal lookback for hedging depends on the severity of the crash. Fast trend following is a low-cost hedge for minor turbulence, while slow trend following is the superior insurance against catastrophic tail events.

9 Conclusion

This paper developed a theoretical framework for understanding trend-following strategies under general stationary Gaussian processes. By decomposing performance into drift and timing components, we have shown that trend returns arise from two distinct sources: the strategy’s average directional exposure to the underlying drift and the alignment between the trend signal and subsequent returns. This decomposition holds both unconditionally and conditionally, providing a unified lens for analyzing trend performance across different market regimes.

Our asymptotic analysis reveals that trend strategies possess robust hedging properties in tail events. For fixed-horizon deep crashes, we established conditions under which trend provides an asymptotically perfect hedge. For sustained linear drawdowns, we proved a threshold property: any stationary Gaussian process with short memory eventually allows profitable trend-following during sufficiently steep declines.

The empirical analysis of 103 assets reveals an asymmetry in the statistical evidence. The theory’s conditional predictions on the convex payoff profile, significant downside betas approaching -1 at matching horizons, and effective hedging across drawdown regimes are confirmed with strong statistical significance. In contrast, the unconditional timing component, while economically positive, does not achieve reliable statistical significance. The robust and reliable value of trend-following lies not in a directional bet on positive autocorrelation, but in the *shape* of the payoff it delivers. The highly significant negative alpha (“cost of convexity”) reinforces this interpretation, positioning trend-following as an insurance-like strategy: it provides a reliable convex payoff and robust tail protection, but at a quantifiable premium much like a put option whose expected return is negative but whose conditional payoff in a crisis is valuable. However, the data do not conclude whether the timing effect is significantly positive. You can rely on the hedging properties in long and severe drawdowns but not on an attractive long term expected return.

Furthermore, our matrix analysis of the lookback windows (k) versus the evaluation horizons (T) uncovers two critical practical insights. First, the “cost of convexity” (annualized alpha) is high for short horizons but declines significantly for longer lookback and holding periods, though for short lookback periods this comes with the price of less beta participation. Second, there is a distinct trade-off in drawdown protection: fast signals ($k = 21$) are superior for hedging shallow mean-reverting corrections, while slow signals ($k = 252$) are essential for capturing the large trends inherent in deep, prolonged crashes.

Several directions remain for future work. Extensions to non-Gaussian processes would broaden applicability, as would analysis of optimal lookback selection and portfolio construction. The Gaussian framework developed here, together with its empirical validation, provides a foundation for understanding the fundamental drivers of trend performance and its role as a portfolio hedge. In particular, our results suggest that trend follow-up is best understood and valued not as a source of unconditional excess returns, but as a provider of convex payoffs and reliable downside protection, complementing the empirical literature on time-series momentum (Moskowitz et al., 2012; Asness et al., 2013) and the practitioner insights on trend follow-up as a risk management tool (Fung and

Hsieh, 2001; Hurst et al., 2017).

A Derivation of Gaussian Half-Space Identities

This appendix provides a detailed derivation of the Gaussian half-space identities stated in Section 3 and used throughout the paper. These identities allow us to compute expectations of the form $\mathbb{E}[X \mathbb{1}\{S > 0\}]$ and $\mathbb{E}[X \text{sign}(S)]$ when (X, S) is bivariate normal.

A.1 Unconditional Results

Setup

Let (X, S) be a bivariate normal random vector with means (m_x, m_s) , variances (v_x, v_s) , and covariance $c_{sx} = \text{Cov}(S, X)$. The correlation coefficient is

$$\rho = \frac{c_{sx}}{\sqrt{v_s v_x}}.$$

The joint density of (X, S) is

$$f_{X,S}(x, s) = \frac{1}{2\pi\sqrt{v_x v_s}(1 - \rho^2)} \exp\left(-\frac{1}{2(1 - \rho^2)} \left[\frac{(x - m_x)^2}{v_x} - 2\rho \frac{(x - m_x)(s - m_s)}{\sqrt{v_x v_s}} + \frac{(s - m_s)^2}{v_s} \right]\right). \quad (41)$$

Derivation of $\mathbb{E}[X \mathbb{1}\{S > 0\}]$

We want to compute

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = \int_{-\infty}^{\infty} \int_0^{\infty} x f_{X,S}(x, s) ds dx.$$

Step 1: Conditional expectation approach. We use the law of iterated expectations:

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = \mathbb{E}[\mathbb{E}[X | S] \mathbb{1}\{S > 0\}]. \quad (42)$$

For bivariate normal (X, S) , the conditional distribution of X given $S = s$ is normal with mean and variance

$$\mathbb{E}[X | S = s] = m_x + \frac{c_{sx}}{v_s}(s - m_s) =: \mu_{X|S}(s), \quad (43)$$

$$\text{Var}(X | S = s) = v_x - \frac{c_{sx}^2}{v_s} = v_x(1 - \rho^2) =: \sigma_{X|S}^2. \quad (44)$$

Step 2: Integrate over $S > 0$. Using (42), we have

$$\begin{aligned}\mathbb{E}[X \mathbb{1}\{S > 0\}] &= \int_0^\infty \mathbb{E}[X \mid S = s] f_S(s) ds \\ &= \int_0^\infty \left(m_x + \frac{c_{sx}}{v_s}(s - m_s) \right) f_S(s) ds,\end{aligned}\tag{45}$$

where $f_S(s) = \frac{1}{\sqrt{2\pi v_s}} \exp\left(-\frac{(s-m_s)^2}{2v_s}\right)$ is the marginal density of $S \sim \mathcal{N}(m_s, v_s)$.

Step 3: Split into two terms. We decompose (45) as

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = m_x \int_0^\infty f_S(s) ds + \frac{c_{sx}}{v_s} \int_0^\infty (s - m_s) f_S(s) ds.$$

The first integral is simply

$$\int_0^\infty f_S(s) ds = \Pr(S > 0) = \Phi\left(\frac{m_s}{\sqrt{v_s}}\right),$$

where Φ is the standard normal CDF.

Step 4: Evaluate the second integral. For the second integral, let $Z = (S - m_s)/\sqrt{v_s} \sim \mathcal{N}(0, 1)$. Then $s - m_s = \sqrt{v_s}Z$ and $ds = \sqrt{v_s} dZ$. The integral becomes

$$\begin{aligned}\int_0^\infty (s - m_s) f_S(s) ds &= \int_{-m_s/\sqrt{v_s}}^\infty \sqrt{v_s}Z \cdot \frac{1}{\sqrt{2\pi}} e^{-Z^2/2} dZ \\ &= \sqrt{v_s} \int_{-m_s/\sqrt{v_s}}^\infty Z \phi(Z) dZ,\end{aligned}\tag{46}$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal density.

Step 5: Standard Gaussian integral. Using the identity $Z\phi(Z) = -\frac{d}{dZ}\phi(Z)$, we have

$$\begin{aligned}\int_{-m_s/\sqrt{v_s}}^\infty Z\phi(Z) dZ &= - \int_{-m_s/\sqrt{v_s}}^\infty \frac{d\phi(Z)}{dZ} dZ \\ &= - [\phi(Z)]_{-m_s/\sqrt{v_s}}^\infty \\ &= - \left(0 - \phi\left(-\frac{m_s}{\sqrt{v_s}}\right) \right) \\ &= \phi\left(\frac{m_s}{\sqrt{v_s}}\right),\end{aligned}\tag{47}$$

where we used $\phi(-z) = \phi(z)$ (symmetry of the standard normal density).

Step 6: Combine results. Substituting (47) into (46), we get

$$\int_0^\infty (s - m_s) f_S(s) ds = \sqrt{v_s} \phi\left(\frac{m_s}{\sqrt{v_s}}\right).$$

Therefore,

$$\mathbb{E}[X \mathbb{1}\{S > 0\}] = m_x \Phi\left(\frac{m_s}{\sqrt{v_s}}\right) + \frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right). \quad (48)$$

Derivation of $\mathbb{E}[X \text{sign}(S)]$

Recall that $\text{sign}(S) = \mathbb{1}\{S > 0\} - \mathbb{1}\{S \leq 0\}$. By symmetry of expectations,

$$\begin{aligned} \mathbb{E}[X \text{sign}(S)] &= \mathbb{E}[X \mathbb{1}\{S > 0\}] - \mathbb{E}[X \mathbb{1}\{S \leq 0\}] \\ &= \mathbb{E}[X \mathbb{1}\{S > 0\}] - (\mathbb{E}[X] - \mathbb{E}[X \mathbb{1}\{S > 0\}]) \\ &= 2\mathbb{E}[X \mathbb{1}\{S > 0\}] - m_x. \end{aligned} \quad (49)$$

Substituting (48) into (49), we obtain

$$\mathbb{E}[X \text{sign}(S)] = 2 \left(m_x \Phi\left(\frac{m_s}{\sqrt{v_s}}\right) + \frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right) \right) - m_x. \quad (50)$$

This can be rewritten more compactly as

$$\mathbb{E}[X \text{sign}(S)] = m_x \left(2\Phi\left(\frac{m_s}{\sqrt{v_s}}\right) - 1 \right) + 2 \frac{c_{sx}}{\sqrt{v_s}} \phi\left(\frac{m_s}{\sqrt{v_s}}\right), \quad (51)$$

which is the form used in Section 4.1 of the main text.

Interpretation

The identity (51) decomposes the expected signed payoff into two components:

1. **Directional component:** $m_x(2\Phi(m_s/\sqrt{v_s}) - 1)$.

The term $2\Phi(m_s/\sqrt{v_s}) - 1 = \mathbb{E}[\text{sign}(S)]$ is the average position (probability of being long minus probability of being short). When multiplied by the unconditional mean m_x , this gives the expected return from simply holding the average position.

2. **Timing component:** $2 \frac{c_{sx}}{\sqrt{v_s}} \phi(m_s/\sqrt{v_s})$.

This term captures the additional profit from timing: being more likely to be long when X is high and short when X is low. The covariance c_{sx} measures this alignment between the signal S and the return X . The Gaussian factor $\phi(m_s/\sqrt{v_s})$ weights this term, with maximum impact when $m_s \approx 0$ (signal mean near the threshold) and diminishing impact as $|m_s|$ grows large (strongly biased signal).

Alternative Derivation via Direct Integration

For completeness, we sketch an alternative direct derivation of (48). Starting from the joint density (41), we can complete the square in x to obtain

$$f_{X,S}(x, s) = f_S(s) \cdot f_{X|S}(x | s),$$

where $f_{X|S}(x | s)$ is the conditional density of X given $S = s$, which is $\mathcal{N}(\mu_{X|S}(s), \sigma_{X|S}^2)$. Then

$$\begin{aligned} \mathbb{E}[X \mathbb{1}\{S > 0\}] &= \int_0^\infty \int_{-\infty}^\infty x f_S(s) f_{X|S}(x | s) dx ds \\ &= \int_0^\infty f_S(s) \left(\int_{-\infty}^\infty x f_{X|S}(x | s) dx \right) ds \\ &= \int_0^\infty f_S(s) \mu_{X|S}(s) ds \\ &= \int_0^\infty f_S(s) \left(m_x + \frac{c_{sx}}{v_s}(s - m_s) \right) ds, \end{aligned} \tag{52}$$

which recovers (45), and the rest of the derivation proceeds as before.

A.2 Conditional Results

The Gaussian half-space identities derived above extend naturally to conditional settings. In the main text (Section 5), we condition on the benchmark cumulative return $B_T = b$ and apply the identities to the conditional distribution of $(X_t, S_t | B_T = b)$. Here we establish the key properties that justify this approach.

Preservation of Normality

Consider the trivariate normal vector (X_t, S_t, B_T) under the stationary Gaussian model. Since X_t , $S_t = \sum_{j=1}^k X_{t-j}$, and $B_T = \sum_{u=1}^T X_u$ are all linear combinations of Gaussian returns $\{X_s\}$, the vector (X_t, S_t, B_T) is jointly normal with mean vector

$$\boldsymbol{\mu} = \begin{pmatrix} \mu \\ k\mu \\ T\mu \end{pmatrix}$$

and covariance matrix

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma^2 & \text{Cov}(X_t, S_t) & \text{Cov}(X_t, B_T) \\ \text{Cov}(S_t, X_t) & \sigma_S^2 & \text{Cov}(S_t, B_T) \\ \text{Cov}(B_T, X_t) & \text{Cov}(B_T, S_t) & \text{Var}(B_T) \end{pmatrix}.$$

By the properties of the multivariate normal distribution, conditioning on $B_T = b$ yields a conditional bivariate normal distribution for $(X_t, S_t \mid B_T = b)$.

Conditional Parameters

The conditional mean vector and covariance matrix of (X_t, S_t) given $B_T = b$ are given by the standard formulas for conditional Gaussian distributions. Let

$$\mathbf{Z} = \begin{pmatrix} X_t \\ S_t \end{pmatrix}, \quad \boldsymbol{\mu}_1 = \begin{pmatrix} \mu \\ k\mu \end{pmatrix}, \quad \boldsymbol{\Sigma}_{11} = \begin{pmatrix} \sigma^2 & \text{Cov}(X_t, S_t) \\ \text{Cov}(S_t, X_t) & \sigma_S^2 \end{pmatrix},$$

and

$$\boldsymbol{\Sigma}_{12} = \begin{pmatrix} \text{Cov}(X_t, B_T) \\ \text{Cov}(S_t, B_T) \end{pmatrix}, \quad \Sigma_{22} = \text{Var}(B_T).$$

The conditional distribution is

$$(\mathbf{Z} \mid B_T = b) \sim \mathcal{N}\left(\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-1}(b - T\mu), \boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-1}\boldsymbol{\Sigma}_{12}^\top\right).$$

Explicit Conditional Moments

Since $\Sigma_{22} = \text{Var}(B_T)$ is scalar, $\Sigma_{22}^{-1} = 1/\text{Var}(B_T)$. The conditional mean vector becomes

$$\mathbb{E}[X_t \mid B_T = b] = \mu + \frac{\text{Cov}(X_t, B_T)}{\text{Var}(B_T)}(b - T\mu) =: m_x(b), \quad (53)$$

$$\mathbb{E}[S_t \mid B_T = b] = k\mu + \frac{\text{Cov}(S_t, B_T)}{\text{Var}(B_T)}(b - T\mu) =: m_s(b). \quad (54)$$

The conditional covariance matrix is

$$\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{12}\Sigma_{22}^{-1}\boldsymbol{\Sigma}_{12}^\top = \begin{pmatrix} \sigma^2 & \text{Cov}(X_t, S_t) \\ \text{Cov}(S_t, X_t) & \sigma_S^2 \end{pmatrix} - \frac{1}{\text{Var}(B_T)} \begin{pmatrix} \text{Cov}(X_t, B_T) \\ \text{Cov}(S_t, B_T) \end{pmatrix} \begin{pmatrix} \text{Cov}(X_t, B_T) & \text{Cov}(S_t, B_T) \end{pmatrix}.$$

Computing each entry:

$$v_x(b) := \text{Var}(X_t \mid B_T = b) = \sigma^2 - \frac{\text{Cov}(X_t, B_T)^2}{\text{Var}(B_T)}, \quad (55)$$

$$v_s(b) := \text{Var}(S_t \mid B_T = b) = \sigma_S^2 - \frac{\text{Cov}(S_t, B_T)^2}{\text{Var}(B_T)}, \quad (56)$$

$$c_{sx}(b) := \text{Cov}(X_t, S_t \mid B_T = b) = \text{Cov}(X_t, S_t) - \frac{\text{Cov}(X_t, B_T)\text{Cov}(S_t, B_T)}{\text{Var}(B_T)}. \quad (57)$$

Note that despite the notation $v_x(b)$, $v_s(b)$, $c_{sx}(b)$, these conditional variances and covariance are actually *constants* (independent of the realized value b) for Gaussian systems—only the conditional means depend on b .

Application of Half-Space Identities

Once we have established that $(X_t, S_t \mid B_T = b)$ is bivariate normal with parameters $(m_x(b), m_s(b), v_x(b), v_s(b), c_{sx}(b))$, the half-space identities derived earlier apply directly with these conditional parameters in place of the unconditional ones. Specifically,

$$\mathbb{E}[X_t \mathbb{1}\{S_t > 0\} \mid B_T = b] = m_x(b) \Phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right) + \frac{c_{sx}(b)}{\sqrt{v_s(b)}} \phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right), \quad (58)$$

$$\begin{aligned} \mathbb{E}[X_t \text{sign}(S_t) \mid B_T = b] &= 2\mathbb{E}[X_t \mathbb{1}\{S_t > 0\} \mid B_T = b] - m_x(b) \\ &= m_x(b) \left(2\Phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right) - 1 \right) + 2 \frac{c_{sx}(b)}{\sqrt{v_s(b)}} \phi\left(\frac{m_s(b)}{\sqrt{v_s(b)}}\right). \end{aligned} \quad (59)$$

This is precisely the form used in equation (20) of Section 5.

Interpretation

The conditional half-space identities preserve the same two-component structure as the unconditional case:

1. **Conditional directional component:** $m_x(b)(2\Phi(m_s(b)/\sqrt{v_s(b)}) - 1)$ represents the conditional expected return $m_x(b)$ times the conditional average position $\mathbb{E}[\text{sign}(S_t) \mid B_T = b]$.
2. **Conditional timing component:** $2c_{sx}(b)/\sqrt{v_s(b)} \cdot \phi(m_s(b)/\sqrt{v_s(b)})$ captures the additional edge from the conditional alignment between signal and return.

The key insight is that conditioning on the cumulative benchmark return $B_T = b$ updates our beliefs about both the expected returns $m_x(b)$ and the signal strength $m_s(b)$, while the conditional volatilities remain well-defined constants. This allows the strategy's conditional performance to be analyzed using the same Gaussian half-space framework as the unconditional case.

B Data Universe

Table 5 lists the 103 assets included in the empirical study, spanning equity indices, government bonds, commodities, and credit/inflation markets.

Table 5: Asset Universe (103 Instruments)

Asset Class	Instruments
Equities (32)	S&P 500, S&P 500 Fut, SPTSX 60 Fut, MSCI Canada, MSCI Mexico, MSCI Mexico Fut, Ibovespa Fut, FTSE 100 Fut, MSCI UK, DAX 50 Fut, MSCI Germany, CAC 40 Fut, MSCI France, FTSE-MIB Fut, MSCI Italy, IBEX 35 Fut, MSCI Spain, SMI Fut, MSCI Swiss, MSCI Denmark, KAX, MSCI South Africa, South Africa Fut, Nikkei 225, Topix Fut, MSCI Japan, KOSPI 200 Fut, Hang Seng Fut, MSCI Hong Kong, MSCI Singapore, SGX MSCI Singapore Fut, MSCI India, Nifty 50 Fut, SPI 200 Fut, MSCI Australia
Govt Bonds (48)	US Treasury (2Y, 3Y, 5Y, 10Y), US Bond Futures (TU, FV, TY), German Bund (2Y, 3Y, 5Y, 10Y), German Bond Futures (DU, OE, RX), UK Gilts (3Y, 5Y, 10Y), UK Gilt Futures (G), French OATs (3Y, 5Y, 10Y), Japan JGB (3Y, 5Y, 10Y), JGB Futures (JB), Australian Bonds (3Y, 5Y, 10Y), AUS Bond Futures (YM, XM), Canadian Bonds (3Y, 5Y, 10Y), Canadian Bond Futures (XQ, CN)
Commodities (13)	Gold Fut, Copper Fut, Aluminum Fut, Nickel Fut, Zinc Fut, Crude Oil (WTI) Fut, Brent Oil Fut, Oil (Spot)
Credit/Inflation (10)	US High Yield, Euro High Yield, Breakeven Inflation (US 7-10Y, UK 7-10Y, Euro/Ger 7-10Y)

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